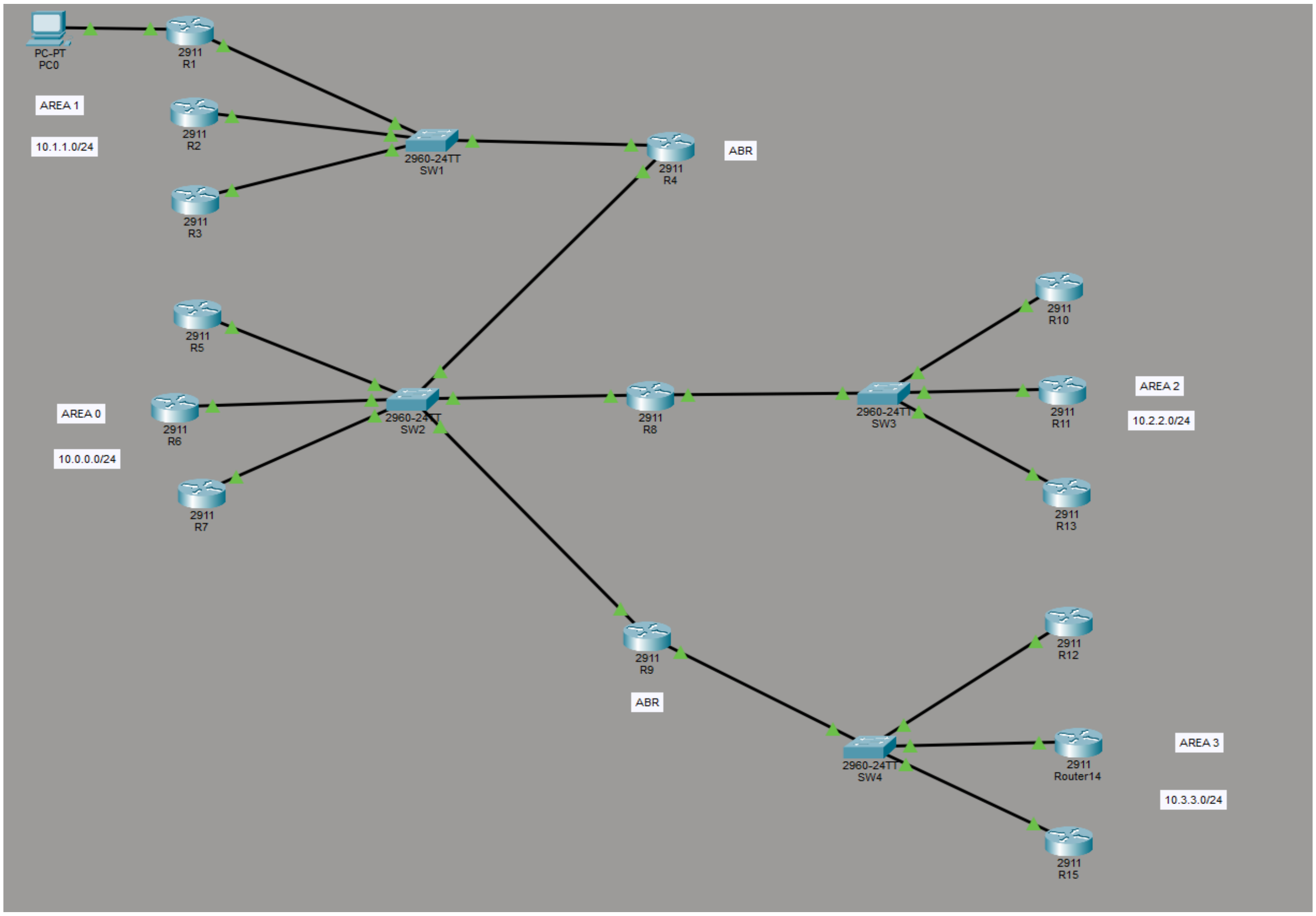
MULTI-AREA OSPF NETWORK

Backbone Area 0 with Area 1, Area 2 & Area 3 (Ethernet Topology)

DIAGRAM:



DESCRIPTION:

This project simulates a multi-area OSPF network made up of one backbone area (Area 0) and three regular areas (Area 1, Area 2 and Area 3) connected to it. Each area contains a LAN switch with a small group of routers attached, representing the end networks of that area. Two routers are configured as ABRs (Area Border Routers), each holding one interface in Area 0 and one interface in a neighboring area, which is the standard way OSPF connects multiple areas back to a single backbone. A third router links the backbone to Area 2 in the same way.

In real-world networks, large multi-area OSPF designs are used to limit the size of the link-state database on any one router, reduce SPF recalculation scope, and allow route summarization at area boundaries. This lab keeps every area small and flat, but it still demonstrates the core ideas: a single backbone area, ABRs connecting non-backbone areas to it, and inter-area routing between all four areas.

STEP 1: IP Addressing Plan (by Area):

Only the LAN network address for each area is shown below, since that is what the topology defines. Point-to-point links between switches/ABRs are not labeled in the diagram, so assign any suitable addressing (e.g. a separate /30 range) for those links when configuring.

Area	Role	Network Address	Devices on the LAN
Area 0	Backbone	10.0.0.0 /24	R5, R6, R7 (via SW2)
Area 1	Regular area	10.1.1.0 /24	R1, R2, R3, PC0 (via SW1)
Area 2	Regular area	10.2.2.0 /24	R10, R11, R13 (via SW3)
Area 3	Regular area	10.3.3.0 /24	R12, Router14, R15 (via SW4)

STEP 2: Configure Area 1 — SW1, R1, R2, R3 & PC0:

SW1 simply needs each router-facing port set as an access port (no VLANs required, since all three routers and PC0 sit on the same Area 1 LAN), plus the uplink port toward the ABR.

SW1 (2960):

interface range fa0/1 - 3 switchport mode access

interface fa0/24

switchport mode trunk (or leave as a normal access/uplink port if no VLANs are used)

R1, R2, R3 (each):

interface Gig0/0

ip address <assign from 10.1.1.0/24> no shutdown

router ospf 1

network 10.1.1.0 0.0.0.255 area 1

PC0 hangs directly off R1 (Gig0/1) and represents an end host inside Area 1 — give it an address from 192.168.1.0/24 with R1's interface as its default gateway.

STEP 3: Configure the Area 0 Backbone — SW2, R5, R6, R7:

SW2 (2960-24TT):

interface range fa0/1 - 3 switchport

mode access

R5, R6, R7 (each):

interface Gig0/0

ip address <assign from 10.0.0.0/24> no shutdown

router ospf 1

network 10.0.0.0 0.0.0.255 area 0

STEP 4: Configure ABR1 (Area 1 <-> Area 0):

This router has one interface toward SW1 (Area 1 side) and one interface toward SW2 (Area 0 side). The interface facing each switch is placed into the OSPF area that side actually belongs to — that is what makes it an ABR.

ABR1:

interface Gig0/0 (toward SW1 / Area 1) ip address <point-to-point address>

no shutdown

interface Gig0/1 (toward SW2 / Area 0) ip address <point-to-point address>

no shutdown

router ospf 1

network <Gig0/0 link network> area 1 network <Gig0/1 link network> area 0

STEP 5: Configure the Link to Area 2 — R8, SW3, R10, R11, R13:

R8 connects the Area 0 backbone (via SW2) to the Area 2 switch (SW3), so it carries the same Area 0 / Area 2 split on its two interfaces as the dedicated ABRs do.

SW3 (2960-24TT):

interface range fa0/1 - 3 switchport mode access

R8:

interface Gig0/0 (toward SW2 / Area 0) ip address <point-to-point address>

no shutdown

interface Gig0/1 (toward SW3 / Area 2) ip address <point-to-point address>

no shutdown

router ospf 1

network <Gig0/0 link network> area 0 network <Gig0/1 link network> area 2

R10, R11, R13 (each):

interface Gig0/0

ip address <assign from 10.2.2.0/24> no shutdown

router ospf 1

network 10.2.2.0 0.0.0.255 area 2

STEP 6: Configure ABR2 (Area 0 <-> Area 3):

ABR2:

interface Gig0/0 (toward SW2 / Area 0) ip address <point-to-point address>

no shutdown

interface Gig0/1 (toward SW4 / Area 3) ip address <point-to-point address>

no shutdown

router ospf 1

network <Gig0/0 link network> area 0 network <Gig0/1 link network> area 3

STEP 7: Configure Area 3 — SW4, R12, Router14, R15:

SW4 (2960-24TT):

interface range fa0/1 - 3 switchport mode access

R12, Router14, R15 (each):

interface Gig0/0

ip address <assign from 10.3.3.0/24> no shutdown

router ospf 1

network 10.3.3.0 0.0.0.255 area 3

STEP 8: OSPF Area Assignment Summary:

Router	Role	Area(s) on its interfaces
R1, R2, R3	Area 1 LAN routers	Area 1
R5, R6, R7	Area 0 (backbone) LAN routers	Area 0
R10, R11, R13	Area 2 LAN routers	Area 2
R12, Router14, R15	Area 3 LAN routers	Area 3
ABR1	Area 1 <-> Area 0 border	Area 1 and Area 0
R8	Area 0 <-> Area 2 border	Area 0 and Area 2
ABR2	Area 0 <-> Area 3 border	Area 0 and Area 3

STEP 9: Verification:

Run on the routers/ABRs as needed:

show ip route

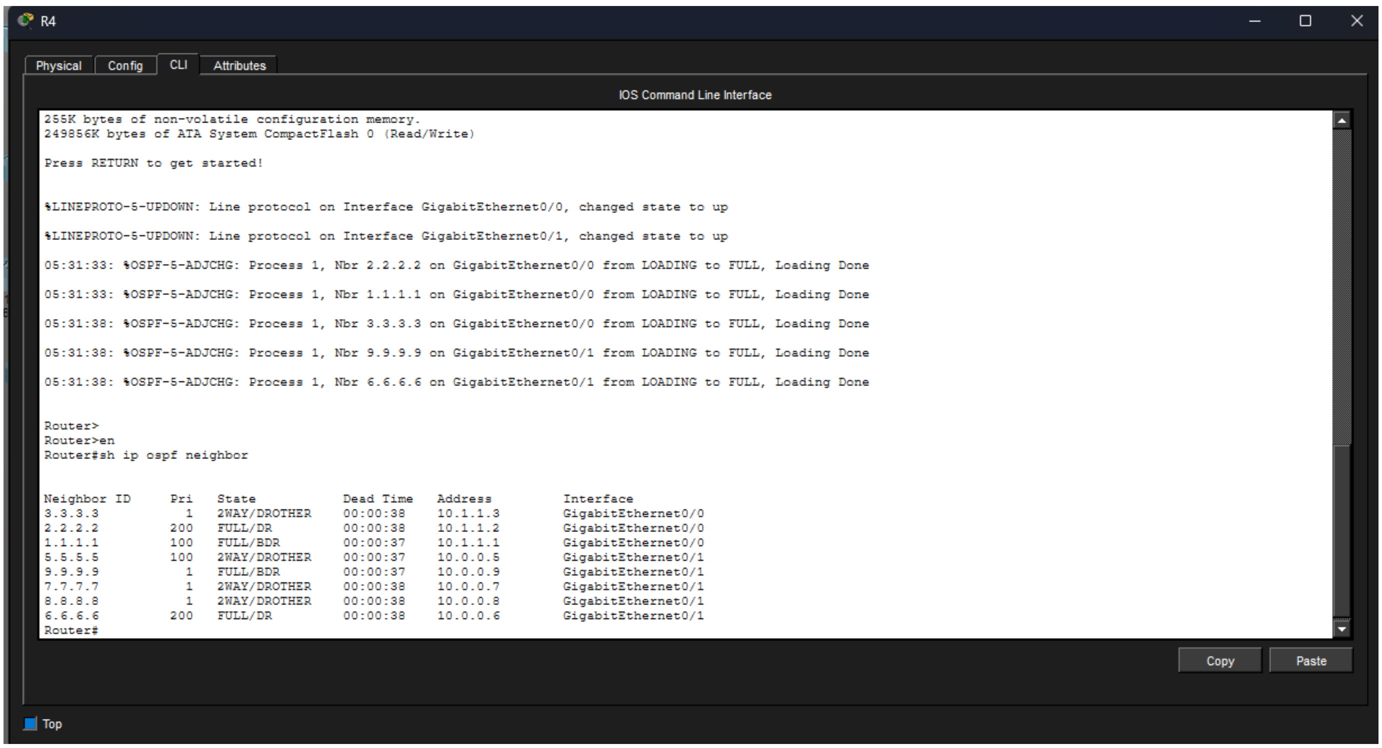
show ip route ospf show ip ospf neighbor

show ip ospf interface brief show ip ospf database

show ip protocols

show ip interface brief

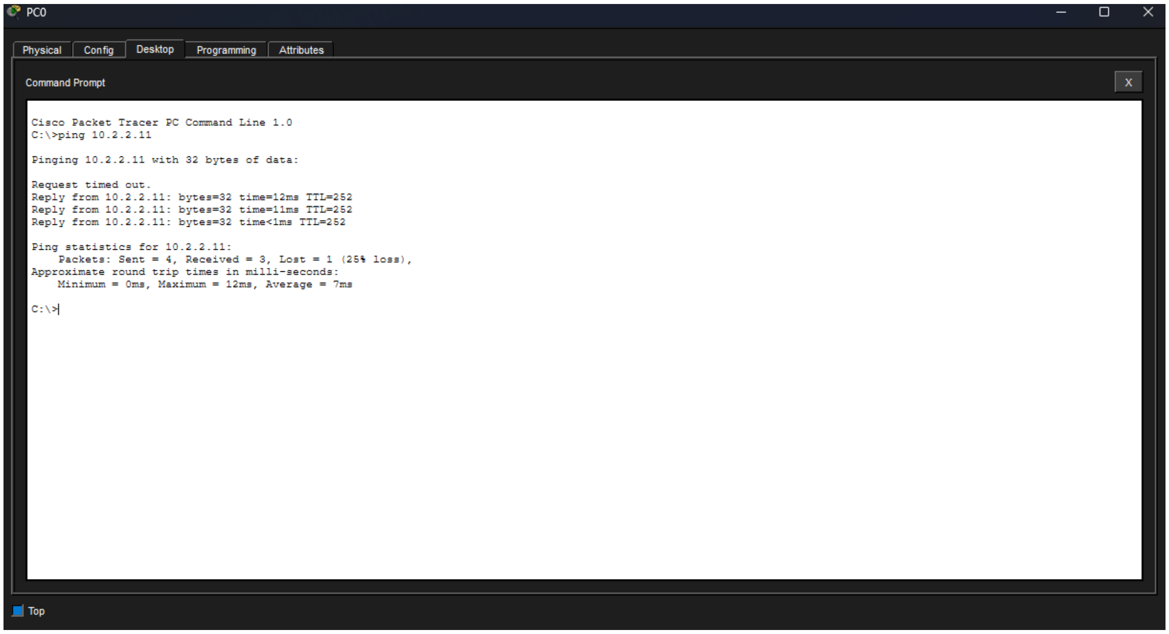
show ip ospf neighbor (ABR1 or ABR2):



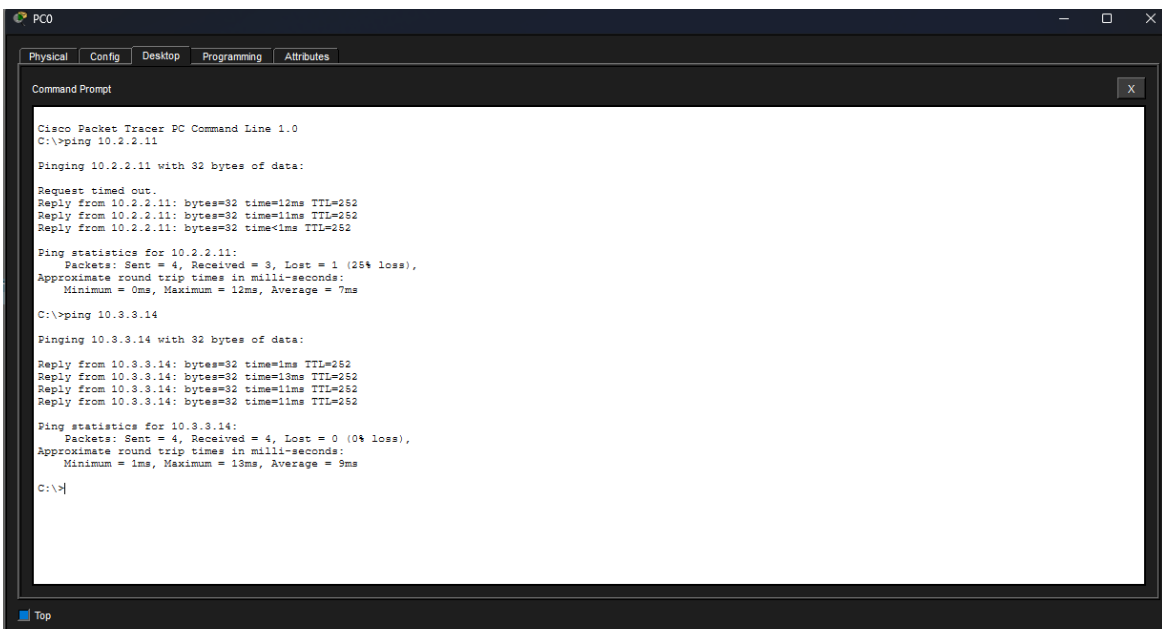
STEP 10: Ping Test:

Verify end-to-end, , for example from PC0 (Area 1) to a host in Area 2 and a host in Area 3.

Ping from PC0 (Area 1) to a host in Area 2 (10.2.2.0/24):



Ping from PC0 (Area 1) to a host in Area 3 (10.3.3.0/24):



NOTE: All four areas (0, 1, 2 and 3) sit on the same internal LAN in this lab for simplicity — the focus here is on multi-area OSPF and ABR behavior rather than WAN transport.